Fiscal Year:	FY 2025	**Task Last Updated:**	FY 02/20/2025
PI Name:	Rice, Kelly Ph.D.		
Project Title:	Assessing the Impact of Agr Quorum Sensing on Staphylococcus aureus Physiology in the Space Flight Environment		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	kcrice@ufl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	352-392-1192
Organization Name:	University of Florida		
PI Address 1:	Microbiology and Cell Science		
PI Address 2:	1355 Museum Dr, Room 1147, PO BOX 110700		
PI Web Page:			
City:	Gainesville	**State:**	FL
Zip Code:	32611-0700	**Congressional District:**	3
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG2. App D: Flight and Ground Space Biology Research
Start Date:	02/08/2021	**End Date:**	02/28/2026
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA KSC
Contact Monitor:	Freeland, Denise	**Contact Phone:**	321-867-5878
Contact Email:	Denise.E.Freeland@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	End date changed to 02/28/2026 per D. Freeland/KSC (Ed., 4/29/25). End date changed to 02/28/2025 per PI information (Ed., 1/8/24).		
Key Personnel Changes/Previous PI:	1. Dr. Wayne Nicholson (Co-I, University of Florida) was removed from this grant in summer 2023, due to his unexpected passing. 2. Dr. Robert Shields (Co-I) is now an Assistant Professor at Arkansas State University. He has been removed as University of Florida (UF) key personnel and we are in the process of setting up a subaward for him to continue his participation on this project. His expertise is required for processing samples and analyzing the TnSeq data to be generated from Aim 1 of this proposal (TnSeq on S. aureus flight and ground control-grown Tn libraries).		
COI Name (Institution):	Carroll, Ronan Ph.D. (Ohio University) Edelmann, Mariola Ph.D. (University of Florida, Gainesville) Shields, Robert Ph.D. (Arkansas State University)		
Grant/Contract No.:	80NSSC21K0601		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Effective protection of astronaut health is predicated on a thorough understanding of the effects of the space flight environment on virulence potential of bacterial pathogens. Staphylococcus aureus is a pathogen that can cause a wide variety of devastating infections, many of which result from the ability of S. aureus to grow as biofilm, its wide arsenal of secreted virulence factors, and its highly adaptive metabolism. These properties are regulated, in part, by its Accessory Gene Regulator (Agr) quorum sensing system. Analysis of S. aureus global gene expression data from a previous space flight experiment (Biological Research In Canisters (BRIC)-23; GeneLab Data System (GLDS)-145)) on the International Space Station (ISS) suggested that the Agr-mediated quorum sensing system (QSS) and expression of some of its target regulon (secreted toxins, exoproteases) were highly upregulated during space flight growth. The overall hypothesis of this proposal is that the S. aureus Agr quorum-sensing system is an important regulator of biofilm formation, physiology, and virulence factor expression when this bacterium is exposed to microgravity or other variables related to the space flight environment. To address this, S. aureus biofilm formation in a wildtype and isogenic agr-deficient mutant strain will be compared using a random positioning machine (RPM) to simulate microgravity, and an ISS space flight experiment will be conducted using 4 Biological Research in Canister Petri Dish Fixation Units (BRIC-PDFUs). One BRIC-PDFU (n=5 PDFUs + 1 HOBO temperature logger) will be used to grow a S. aureus transposon library, in order to define which gene(s) are required for thriving in the space flight environment. Two BRIC-PDFUs will be used to grow S. aureus wildtype (same strain used in BRIC-23) and its isogenic agr mutant (n=5 PDFUs + 1 HOBO temperature logger per strain), to determine the contribution of Agr quorum sensing to gene expression changes previously observed in the BRIC-23 flight samples. In a fourth BRIC, PDFUs modified to cancel the potential confounding effects of microgravity on fluid geometry, and hence aeration, in BRIC units (modifications are being developed by Co-Investigator Nicholson through a separate NASA project), will be used to grow the wild-type S. aureus strain (n=5 modified PDFUs + 1 HOBO temperature). The growth media, time, and other experimental variables will be identical to those previously used in the BRIC-23 experiment in order to directly compare results. Asynchronous ground controls for each of the four space flight BRIC-PDFUs (n=5 PDFUs + 1 HOBO temperature logger per BRIC) described above will be treated identically in the ISS Environmental Simulator at Kennedy Space Center (KSC) using the recorded flight temperature data. Upon return to Earth, flight samples and ground-based controls for the wild-type and agr mutant BRIC-PDFUs will be processed for RNA-seq, secreted proteomics, metabolite, and growth measurements. The transposon library BRIC-PDFUs will be processed for DNA isolation, Illumina sequencing, and Tn-seq analysis. These proposed experiments will allow us to make more refined predictions about the pathogenic potential of S. aureus, especially in relation to astronaut health. These experiments align with Decadal Survey Recommendation P1, sub-topics MB-B (Studies designed to determine how spaceflight conditions affect microbial biofilm formation... and their potential to cause disease) and MB-D (to use the ISS as a microbial observatory to determine foundational results... that contribute to the health of future space explorers), as well as address several guiding questions of the 2016-2025 SB Science Plan: To help identify the "underlying genetic, molecular and biochemical processes influenced by the space environment" for S. aureus (MB-1), and offer additional insight into how space flight conditions such as microgravity "...influence microbial biofilm formation and microbial community development" (MB-3).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Agr is a global regulator of both virulence and metabolic genes, therefore stimulation of this quorum-sensing system in response to space flight conditions may impact the pathogenic threat of S. aureus to astronauts during long-term space flight missions, especially since space flight is known to impose a suppressive effect on both innate and adaptive immune cell responses to pathogen-related stimuli. The research proposed in this application is directed at gaining a better understanding of the contribution of Agr quorum sensing to physiological adaptation and pathogenic potential of S. aureus in response to the space flight environment. It is anticipated that the results of these proposed experiments will allow us to make more refined predictions as to which virulence properties and physiological changes are likely to occur in S. aureus during space flight, and how this may impact its pathogenic potential and effect on astronaut health.
	Aim 1: ISS flight experiment to elucidate the physiological response of S. aureus to the space flight environment A) In year 4, our Biological Research In Canisters (BRIC)-25 flight (FLT) experiment launched on NG-20 on January 30, 2024. Samples were subsequently returned from the International Space Station (ISS), deintegrated, and stored at -80 C. HOBO data logger temperature measurements from the BRIC-25 FLT experiment were used to generate a temperature profile to conduct the corresponding ground control (GC) experiment. Samples from both FLT and GC experiments were subsequently processed for growth measurements (total CFU, OD660, pH), and cell pellets and/or supernatants were stored at -80 C for RNA extraction/RNASeq, metabolomics, secretomics, and DNA extraction/TnSeq. B) “Gizmo” petri dishes, designed to mitigate the influence of microgravity on liquid geometry in BRIC canisters (1), were tested as part of the BRIC-25 experimental design. In previous BRIC flight experiments, liquid cultures displayed a toroidal (“donut”) shape in regular petri dishes compared with the ground control samples (a flat disc shape). Consistent with this, FLT regular petri dish liquid cultures displayed a donut-like shape and GC cultures maintained a flat disc shape. Although gizmo FLT liquid cultures did maintain a flat disc-like shape relative to the donut shape observed in petri FLT liquid cultures, the gizmo plate FLT liquid cultures typically only covered ~1/3 to 2/3 of the total surface area of the gizmo plate. As expected, the gizmo GC2 liquid cultures maintained a disc shape that covered the entire plate. C) Success criteria with respect to FLT and GC sample growth (total recovered CFU) were met for UAMS-1 petri, agr petri, and TnSeq samples. Although Science Verification Test (SVT) and Experiment Verification Test (EVT) experiments did not previously indicate any growth issues with S. aureus on the gizmo petri dishes, surprisingly the gizmo FLT samples displayed greatly reduced viability compared to the other FLT samples. In fact, these cultures were less viable than the initial inoculum. Additionally, GC gizmo culture CFU counts were comparable to the initial inoculum, suggesting that these cultures did not undergo any growth. We are currently investigating theories as to why the FLT and GC gizmo samples did not grow well. D) Secretomics, metabolite detection (acetate, lactate, glucose, glycerol) of BRIC-25 FLT and GC supernatants are currently in progress. The new graduate student working on this project has optimized the secreted protein precipitation protocol using practice samples and will work with Dr. Ferraro (Co-Investigator) to prepare FLT and GC samples using this method, followed by mass spectrometry.

Task Progress:	Carroll subaward progress report (Ohio University): FLT and GC petri cell pellets of wildtype (UAMS-1) and agr mutant samples have been shipped to Dr. Carroll's lab (Co-Investigator, Ohio University), RNA has been isolated, and RNA sequencing (RNASeq) is n progress. We did not ship UAMS-1 gizmo cell pellets (originally planned in our grant) as these FLT and GC samples grew unexpectedly poorly. Shields subaward progress report (Arkansas State University): The new graduate student working on this project is optimizing the genomic DNA extraction protocol (used previously for EVT samples) using practice samples and will work with Dr. Shields (Co-Investigator) to prepare FLT and GC samples using this method, followed by DNA sequencing and TnSeq data analysis. Aim 2: Determine the effect of Agr and simulated microgravity on S. aureus biofilm growth using a random positioning machine (RPM) A) UAMS-1 (wildtype; WT) and agr mutant biofilms were grown at 1 g and 0 g (simulated microgravity using the random positioning machine and previously-published methodology), followed by confocal microscopy and COMSTAT2 analysis to quantify biofilm characteristics. Both WT and Δagr biofilms grown in simulated microgravity displayed significantly less biomass and surface area compared to their respective 1 g controls. Biofilm assays designed to quantify the ratio of attached cells to planktonic cells in these biofilms have been optimized and this experiment will be conducted during our next visit to the Microgravity Simulation Support Facility (MSSF) at NASA Kennedy Space Center (KSC) in February or April 2025. B) As described in last year's progress report, we showed that S. aureus biofilms grown in medium that physiologically represses induction of the Agr quorum sensing system, are able to induce high levels of Agr activity when grown on a random positioning machine (RPM) at 0 g. Therefore, a signal in the 0 g simulation appears to override the Agr repression that occurs in this media in biofilms grown at 1 g. To delve more into this mechanism, planktonic cultures of UAMS-1 were cultured in biofilm medium in HARV vessels on the RPM (0 g) and in static (1 g) control HARVs (n=4 each), followed by RNA isolation and RNAseq. Although the upregulation of a few Agr-regulated genes were detected in the differential expression (DE) analysis of this data, strong induction of RNAIII (whose promoter was strongly upregulated in 0 g RPM biofilms) was not observed. This suggests that altered Agr activity in 0 g RPM cultures may be specific to biofilm growth. Follow-up studies with RNA isolated from RPM biofilms will be undertaken during this next grant period. C) To ascertain the similarity of transcriptome comparisons between 0 g RPM/1 g growth, BRIC-23 FLT/GC and BRIC-25 FLT/GC, S. aureus UAMS-1 was grown in the same medium (TSYG), time point (48 hrs) and approximate temperature (room temperature, ~24 C) in HARVs using the RPM (0 g) or inverted (oxygenation membrane side up) sitting statically (1 g control) (n=4 biological replicates per condition). RNA was isolated from all samples and subjected to RNAseq. Data analysis is currently in progress.
Bibliography Type:	Description: (Last Updated: 02/19/2026)
Articles in Peer-reviewed Journals	Hauserman MR, Sullivan LE, James KL, Ferraro MJ, Rice KC. "Response of Staphylococcus aureus physiology and Agr quorum sensing to low shear modelled microgravity." J. Bacteriology. 2024 Aug 9;206(9):e0027224. https://doi.org/10.1128/jb.00272-24 ; PMID: 39120147 , Aug-2024
Articles in Peer-reviewed Journals	Ocampo J, Barker H, Rice KC, Ferraro MJ. "Impact of payload shielding on Enterobacter cloacae viability and proteomic profile: Insights from a stratospheric weather balloon flight experiment." Gravit Space Res. 2024 Jun 9;12(1):64-76. https://doi.org/10.2478/gsr-2024-0005 , Jun-2024
Articles in Peer-reviewed Journals	Ocampo J, White RE, Ferraro MJ, Rice KC. "Impact of simulated microgravity on the growth and proteomic profile of Enterobacter cloacae." Microbiol Spectr. 2025 Jun 3;13(6):e0244624. https://doi.org/10.1128/spectrum.02446-24 ; PMID: 40272176 PMCID: PMC12131722 , Jun-2025